

MuLERMOCs — Project Plan

Multiple-Level External Representations for Molecular Concepts

Organic Chemistry Interactive Learning & Research Platform

Principal Investigator: N. Charistos, Department of Chemistry, ΑΠΘ **Status:** Active development | v37 stable and deployed **Last updated:** 2026-05-04

Vision

Build a theoretically grounded, research-instrumented interactive platform for organic chemistry learning — covering nomenclature, isomerism, and reactions — that serves both Greek chemistry students and the international chemistry education research community.

The platform generates Greek IUPAC names algorithmically, displays molecules in multiple representations (2D condensed, expanded, annotated skeletal, skeletal, 3D), and supports interactive quiz scenarios designed for both learning and research data collection.

What Makes This Different

Feature	Commercial tools	chemtube3d.com	MuLERMOCs
Algorithmic IUPAC naming	No	No	Yes
Multiple 2D representation modes	No	No	Yes
Synchronized 2D + 3D highlighting	No	No	Yes
Greek language	No	No	Yes
Research instrumentation (logging, condition lock)	No	No	Yes
Quiz interaction scenarios	No	No	Yes
Quantum-accurate IRC trajectories (reactions)	No	Partial	Planned

Theoretical Framework

CTML (Mayer, 2009) — Cognitive Theory of Multimedia Learning

- Guides when multiple representations help vs. hurt learning
- Spatial contiguity principle → synchronized highlighting design

DeFT (Ainsworth, 1999, 2006) — Design, Functions, Tasks framework

- Three functions of multiple representations: complementary, constraining, constructive

- Constraining function → the representation fading sequence (core original contribution)

The Representation Fading Sequence *(original design principle)*

Condensed → Expanded → Annotated Skeletal → Pure Skeletal → 3D
CH3CH2OH (explicit H) (C visible on zig-zag geometry) (C implicit) (spatial geometry)
(textbook)

Each step removes one scaffolding layer. Familiar representations constrain interpretation of unfamiliar ones.

Not previously proposed or tested in chemistry education literature.

Project Modules

MODULE 1: Organic Nomenclature	[ACTIVE – v37 stable]
MODULE 2: Isomerism	[PLANNED – Phase 3]
MODULE 3: Organic Reactions	[PLANNED – Phase 4]

↓ All modules share the same JS engine, viewers, and quiz infrastructure

Module 1: Organic Nomenclature — Current State (v37)

What exists

- **58 curated molecules** covering 9 IUPAC rule groups (hydrocarbons through mixed functional groups)
- **Algorithmic IUPAC engine** — DFS main chain determination, 3-tier locant rules, Greek euphony
- **5 representation modes** — condensed, expanded, annotated skeletal (partial), skeletal, 3D
- **Synchronized highlighting** — same atom/bond highlighted simultaneously in 2D and 3D
- **Teaching layer** — rule-based explanations, component name boxes, TTS narration in Greek
- **Technical validation** — algorithm verified against all 58 hand-curated molecules

Known gaps to fill (immediate work)

- Annotated skeletal representation: `fMakeAnnotatedSkeletal()` — completes the fading sequence
- Atom click targets: `fAddAtomClickTargets()` — enables interactive quiz scenarios
- Branched molecule naming (alkyl substituents) — extends engine to full organic curriculum
- Ether naming (moderate complexity)
- Ester naming (highest complexity — two-chain naming)

Development Roadmap

Phase 1 — Complete Nomenclature Tool + First Quiz Scenario

Timeline: ~3 months

- ☐ `fMakeAnnotatedSkeletal()` → annotated skeletal representation complete
- ☐ `fAddAtomClickTargets()` → atom/bond click infrastructure on 2D viewer
- ☐ JSmol click callback → atom/bond selection on 3D viewer
- ☐ Quiz Scenario A or D → one working click-based identification scenario
- ☐ Think-aloud pilot study → 6–8 students, one session each

Deliverable: Complete fading sequence + first interactive quiz scenario + pilot data

Phase 2 — Full Quiz SPA + Research Mode + First MSc Studies

Timeline: ~12 months

- ☐ Full quiz SPA (`quiz-nom-1.html`) with all 5 scenarios:
 - **A** — Component identification: click atoms matching a named component
 - **B** — Name assembly: arrange shuffled tiles in correct order
 - **C** — Multiple choice: select correct name from 4 options
 - **D** — Reverse direction: given name component, identify atoms
 - **E** — Timed challenge: rapid-fire with countdown
- ☐ Session logging (`mulermoc-nom-log-1.js`) — per-attempt schema, localStorage, CSV export
- ☐ Research mode — condition lock, participant ID, no-feedback option, viewer visibility control
- ☐ Branched molecule naming (alkyl substituents)
- ☐ Ether naming
- ☐ Expanded molecule database (target: 100+ molecules)
- ☐ Transfer-test molecule set (separate from study set — required for valid retention tests)
- ☐ Ethics approval (GDPR compliance for student interaction logging)
- ☐ **MSc Study 1** — Which 2D representation? (condensed vs expanded vs skeletal, between-subjects)
- ☐ **MSc Study 2** — Simultaneous vs single representation? (2D only vs 3D only vs 2D+3D)

Deliverable: Full research-instrumented quiz platform + first two MSc studies underway

Phase 3 — Platform Expansion + PhD Research

Timeline: ~24 months

- ☐ Ester naming
- ☐ Extended molecule database (target: 150–200 molecules, full secondary school curriculum)
- ☐ **Error detection** quiz scenario — identify which name component is wrong
- ☐ **Comparison** and **classification** quiz scenarios
- ☐ Gamification layer (points, adaptive sequencing) — togglable, off in research mode
- ☐ **MSc Study 3** — Interaction type comparison (MC vs click vs build vs error detection)
- ☐ **PhD research** — eye-tracking + think-aloud, representation integration strategies
- ☐ Isomerism module — constitutional and stereoisomerism, E/Z, chirality (shared engine)
- ☐ **Dedicated Drupal platform** — mulermocs.chem.auth.gr (standalone site, ΑΠΘ Chemistry domain)
 - Teacher dashboard: molecule library, filter by class/rule/complexity
 - Teacher quiz builder: select molecules + lock representation condition
 - Generates shareable quiz link for students

- Student interaction data exportable per session
- User accounts, class management

Deliverable: Full platform live at mulermocs.chem.auth.gr + PhD program + 4–6 publications

Phase 4 — Organic Reactions Module

Timeline: 3–5 years

- ☐ Reaction database: simple organic reactions (SN2, E2, addition to carbonyl, etc.)
- ☐ Gaussian IRC calculations → multi-frame trajectory files (offline, PI expertise)
- ☐ JSmol 3D reaction animation (curly arrows natively supported by JSmol)
- ☐ Hand-drawn SVG 2D mechanism diagrams + JSmol 3D synchronized (*model proven by chemtube3d.com, University of Liverpool*)
- ☐ Quiz scenarios for reactions (identify nucleophile, predict stereochemistry, select mechanism step)
- ☐ **Research: reaction mechanism learning** — does Gaussian-accurate IRC animation improve understanding vs. static arrow-pushing?

Differentiator from chemtube3d.com: Research instrumentation, quiz interaction scenarios, quantum-accurate IRC geometries, full curriculum integration (nomenclature → isomerism → reactions), Greek language.

Research Program

Three Independent Research Axes

Axis 1: REPRESENTATION TYPE GAMIFICATION	Axis 2: INTERACTION TYPE	Axis 3:
RQ1: 2D vs 3D gamification	RQ5: Which interaction type	RQ6: Does
RQ2: Which 2D representation? improve outcomes?	leads to best retention?	
RQ3: Simultaneous vs single? badges /	MC → click → build → error	Points /
RQ4: Learner strategies sequencing (eye-tracking)	detection	adaptive
quiz	(testing effect / desirable difficulties predictions)	vs plain
MSc 1, MSc 2, PhD study	MSc 3 / PhD	Future

Study Designs

- **RQ1, RQ2, RQ3:** Between-subjects (each participant sees only one representation condition)
- **RQ4:** Within-session observational — eye-tracking + think-aloud
- **RQ5:** Between-subjects (interaction type as condition)

- All studies: pre-test (prior knowledge + spatial ability) + immediate post-test + 1-week delayed retention

Target Journals

Study	Target journal
DBR tool paper	Chemistry Education Research and Practice (CERP)
RQ2 (which 2D?)	Journal of Chemical Education (JChemEd)
RQ3 (simultaneous vs single)	Computers & Education
RQ4 (eye-tracking strategies)	Learning and Instruction
RQ5 (interaction type)	CERP or JChemEd
Reactions module paper	JChemEd

Estimated output: 6–8 peer-reviewed publications over 5–7 years

Team & Roles

Role	Contribution
PI (N. Charistos)	Engine development, molecule database, research design, Gaussian IRC calculations
Drupal collaborator	Platform integration, mulermocs.chem.auth.gr deployment, teacher quiz builder
MSc Student 1	RQ2 study (2D representation comparison) — thesis
MSc Student 2	RQ3 study (simultaneous vs single) — thesis
PhD Candidate	RQ4 (eye-tracking + think-aloud) + secondary RQs — dissertation
MSc Student 3 (future)	RQ5 (interaction type) — thesis

First Publication: DBR Tool Paper

Title (draft): "Designing for Multiple Representations in Organic Nomenclature Learning: A Design-Based Research Approach" **Target:** Chemistry Education Research and Practice (CERP) **Length:** ~7,500 words

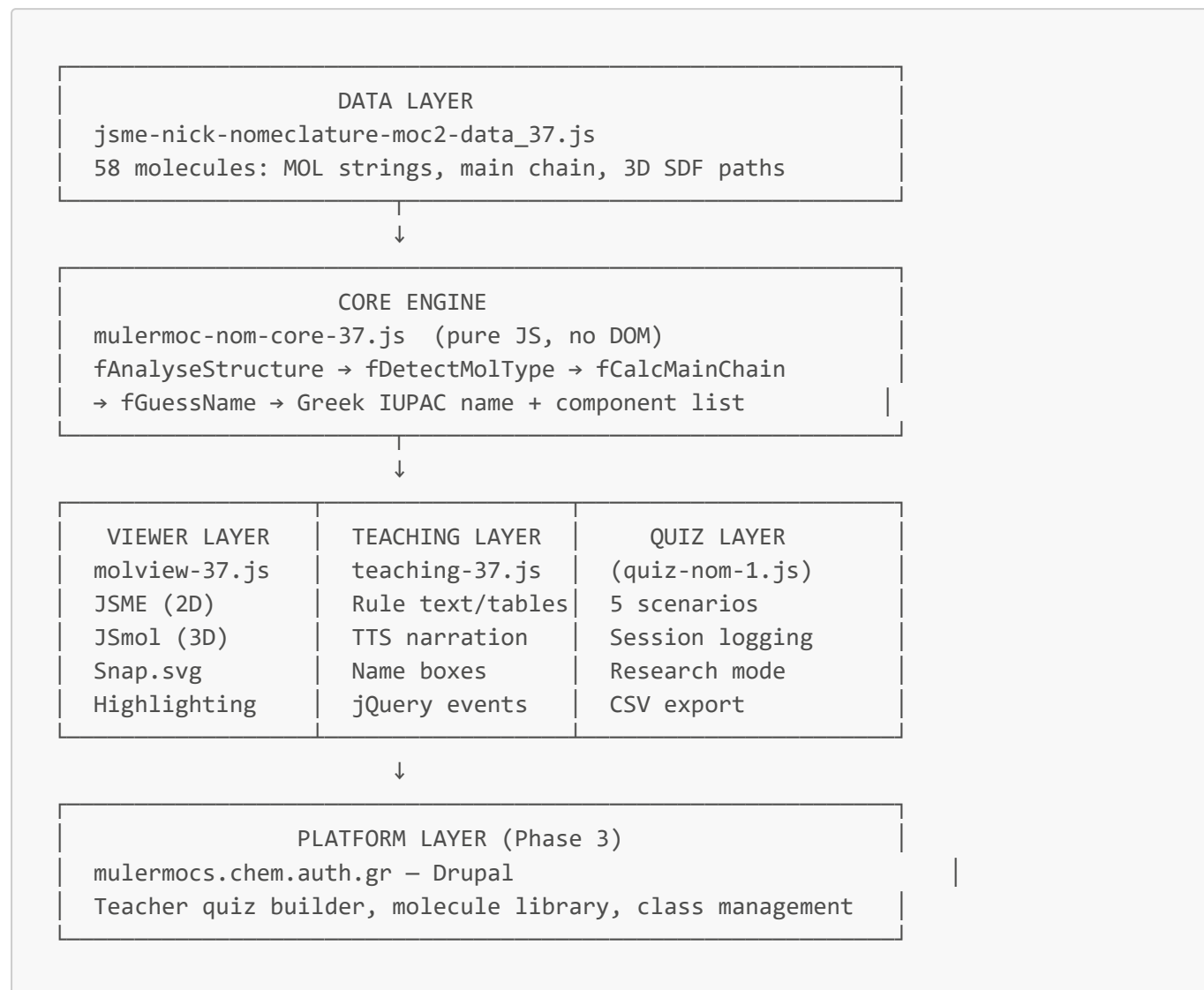
Key contribution: The representation fading sequence as a novel, DeFT-derived pedagogical design principle — not previously proposed or empirically tested in chemistry education.

Prerequisites before submission:

1. `fMakeAnnotatedSkeletal()` implemented (*completes fading sequence*)
2. One click-based quiz scenario working (*demonstrates research instrument*)
3. Think-aloud pilot data collected and coded (*qualitative evidence of tool use*)

Timeline: 3–4 months from pilot data collection to submission

Technical Architecture



Immediate Next Steps

1. Implement `fMakeAnnotatedSkeletal()` in molview — completes the 5-step fading sequence
2. Implement `fAddAtomClickTargets()` — atom/bond click infrastructure
3. Build one click-based quiz scenario (Scenario A: click atoms matching a name component)
4. Run think-aloud pilot with 6–8 students
5. Write DBR tool paper — submit to CERP

Representations for Molecular Concepts